

PE INTERNATIONAL
SUSTAINABILITY PERFORMANCE

Flick It or Ship It

Comparative Life Cycle Assessment of Cigarette Waste Disposal Routes with a Focus on Toxicity Impacts

Trisha Montalbo, PE INTERNATIONAL
Christoph Koffler, PE INTERNATIONAL
Takuma Ono, PE INTERNATIONAL
Ernel Simpson, TerraCycle

Cigarette Waste

Challenges and Concerns



Marine Debris

Animal Ingestion

Toxic



Fire Hazard

Roadside Litter

65% littering rate in the U.S.

US CDC, http://www.cdc.gov/tobacco/
Health Canada, http://www.hc-sc.gc.ca/hc-ps/tobac-tabac/index-eng.php
Keep America Beautiful, http://www.kab.org

Addressing the Problem of Cigarette Waste

TerraCycle's Cigarette Brigade



- Cellulose acetate fibers (from filters) is compounded with recycled polypropylene to produce plastic products
- Paper and tobacco are composted to produce fertilizer



TerraCycle, http://www.terracycle.com/en-US/brigades/cigarette-waste-brigade.html

Evaluate Environmental Costs and Benefits

Application of Life Cycle Assessment (LCA)

QUESTION:
How does recycling cigarette waste compare to alternative disposal routes?



APPROACH:
Apply LCA framework to quantify environmental performance

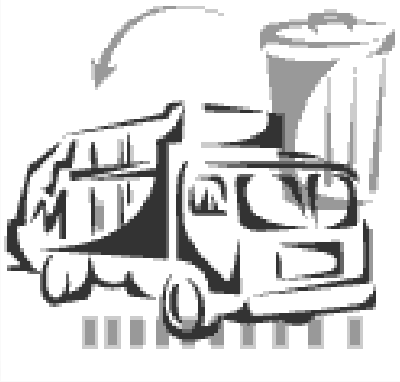
Emphasize toxicity

Product Systems Under Study

Recycling



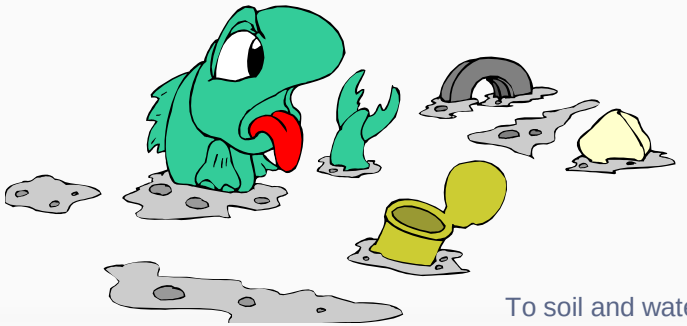
Landfill



Incineration



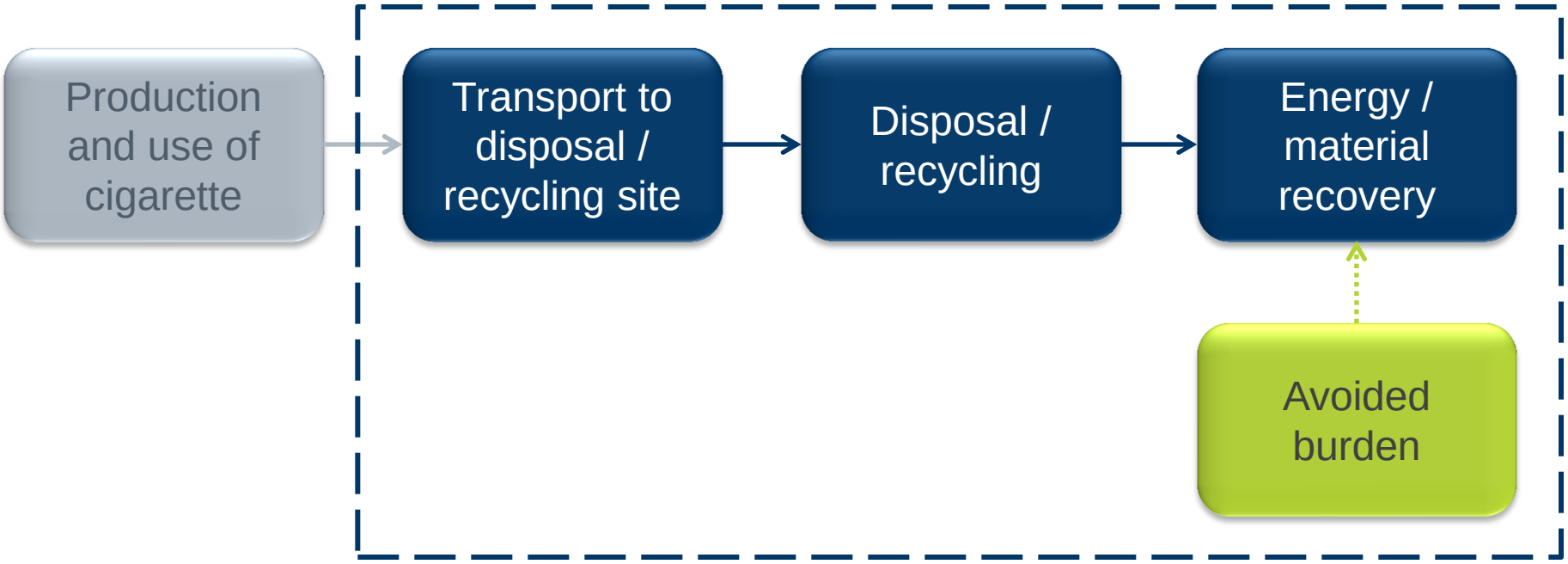
Unmanaged Waste (litter)



Defining the LCA

Gate-to-Grave System Boundaries

Functional unit: The treatment of 1 kg of post-consumer cigarette butts disposed in Canada.



Life Cycle Impact Methodologies

TRACI 2.1, PED, and USEtox

LCIA Impact Category	Abbrev.	Units
Acidification potential	AP	kg SO_2 eq.
Eutrophication potential	EP	kg N eq.
Global warming potential	GWP	kg CO_2 eq.
Ozone depletion potential	ODP	kg CFC 11 eq.
Smog formation potential	SFP	kg O_3 eq.
Primary energy demand, non-renewable	PED	MJ
Freshwater eco-toxicity potential	ETP	CTU_{eco}
Human toxicity potential, cancer	HTPC	CTU_h
Human toxicity potential, non-cancer	HTPNC	CTU_h

CTU: Comparative Toxic Unit









Life Cycle Inventory Model

Tobacco and Filter Toxins

Toxins found in the tobacco
(based on literature)



and in the post-consumer filter
(measured by TerraCycle)

- Acrolein
- Ammonia
- Antimony
- Arsenic
- Barium
- Benzene
- Benzo(a)pyrene
- Cadmium
- Catechol
- Chromium
- Cresol
- DDT
- Formaldehyde
- Hydrogen cyanide
- Hydroquinone
- Isoprene
- Lead
- Mercury
- Methanol
- Nicotine
- NNK
- Phenol
- Resorcinol
- Selenium
- Styrene
- …

Selected toxins defined / provided by:

- Consumer Product Safety Improvement Act of 2008
- Food and Drug Administration's Indirect Food Additives standard (21 CFR – 177.1520)
- Data collected by Health Canada under the Tobacco Act

Life Cycle Inventory Model

Toxin Capture and Emissions

Recycling

- Cellulose acetate filter toxins trapped in recycled plastic
- Tobacco toxins leave system with compost

Landfill

- Majority of toxins trapped in landfill
- Remainder emitted to soil (through leachate) or to wastewater

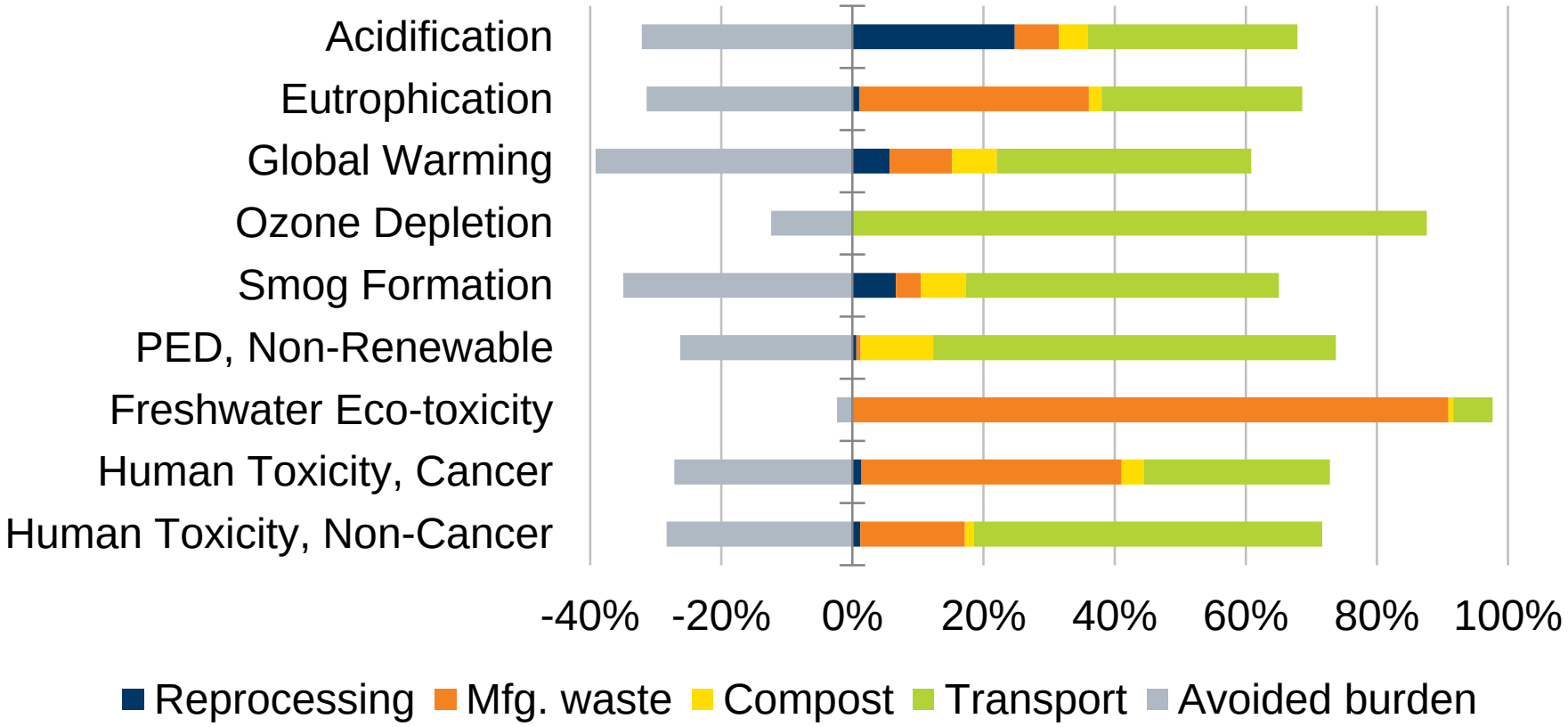
Incineration

- Organic toxins combusted
- Inorganic toxins emitted to atmosphere and/or scrubbed from flue gas

Unmanaged Waste (litter)

- Scenario 1: Emitted to soil
- Scenario 2: Emitted to water

LCIA Results: Recycling

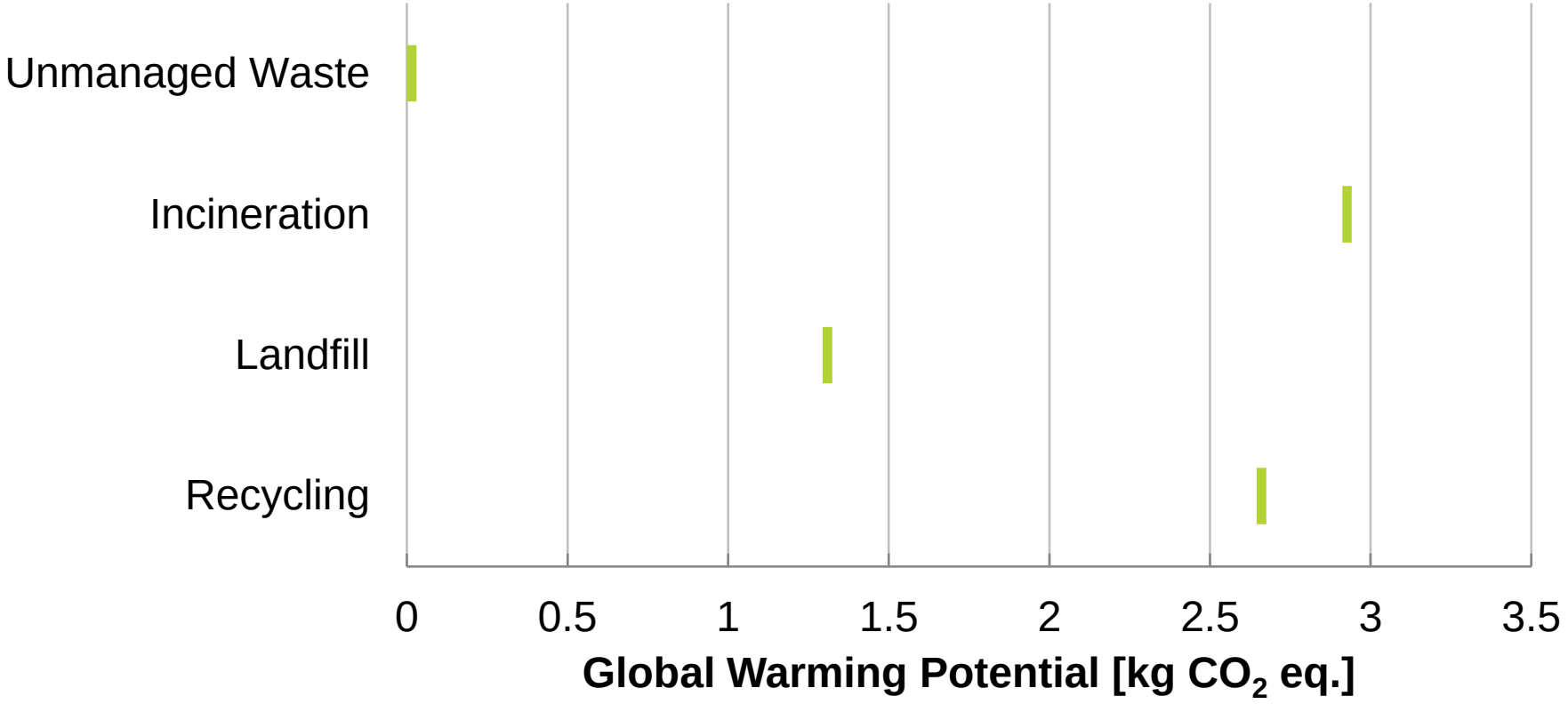


- Impact dominated by transport and packaging of cigarette waste
- Credit from the avoided burden of fertilizer and recycled polypropylene production

Primary Energy Demand, Non-Renewable

Disposal System Impact Comparison

Unmanaged waste not associated with energy consumption



Understanding Toxicity: USEtox™ Model

1) Identify which toxins drive impact

2) Compare toxicity performance

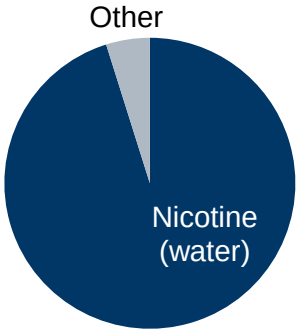
Obtain characterization factors from the USEtox model

- Represents improvement over other toxicity models
- More appropriate for organic toxins than for inorganic toxins (e.g., heavy metals)
- Account for imprecision of characterization factors in interpretation
  - Human toxicity factors: Precision within 100 – 1,000 of other models
  - Eco-toxicity factors: Precision within 10 – 100 of other models

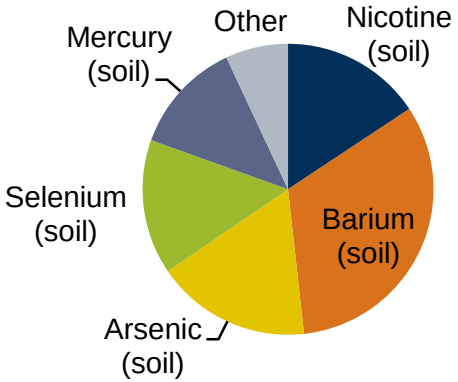
Freshwater Eco-Toxicity Potential

Contributing Toxins

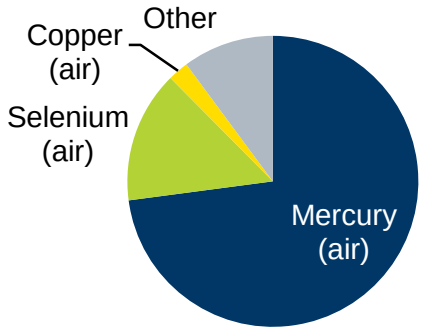
Landfill / Recycling



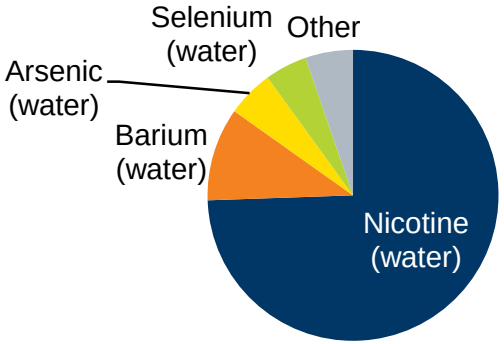
Unmanaged Waste (soil)



Incineration

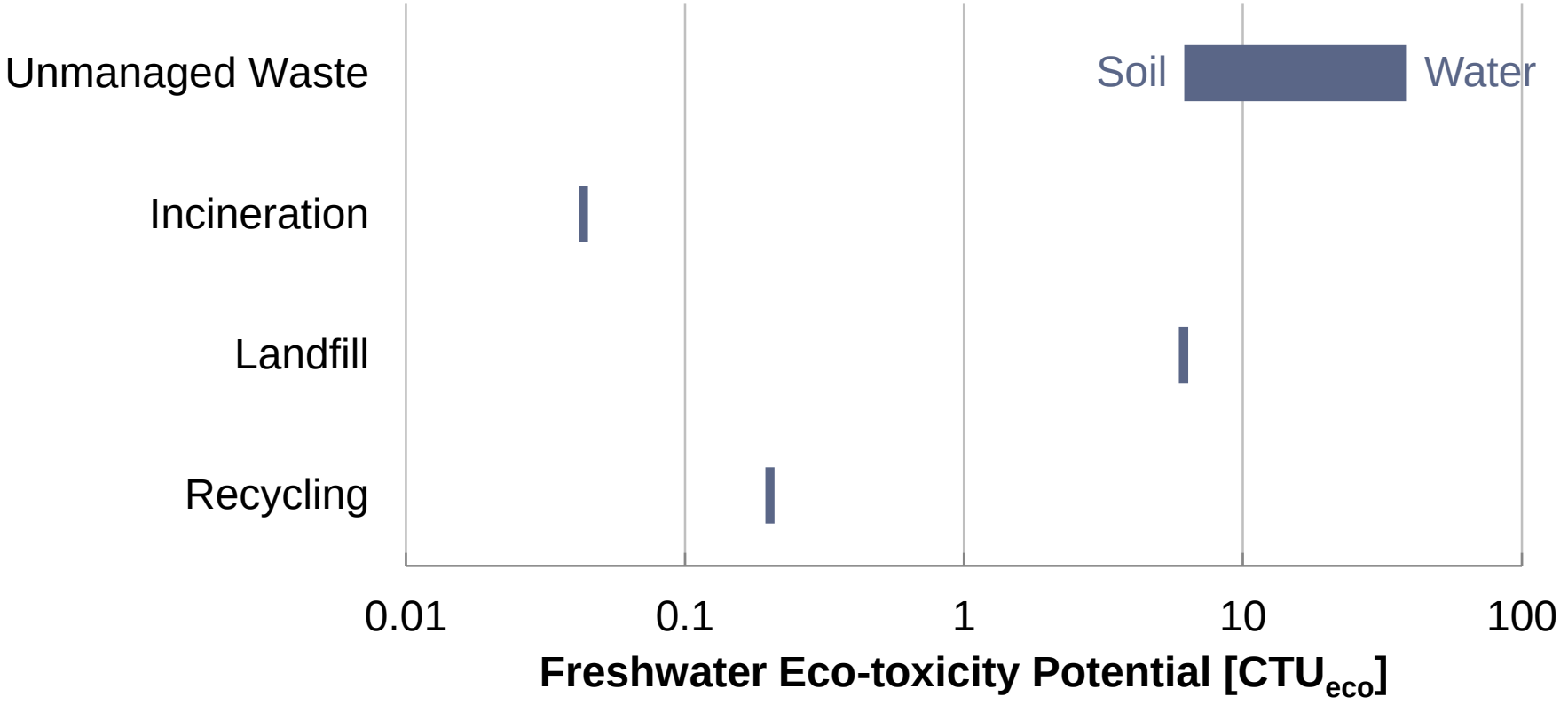


Unmanaged Waste (water)



Freshwater Eco-toxicity Potential

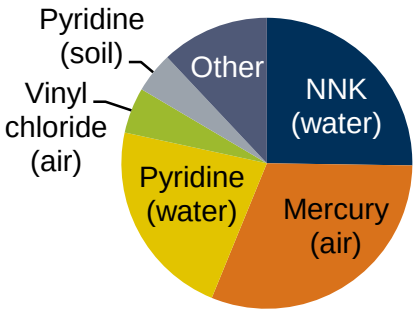
Disposal System Impact Comparison



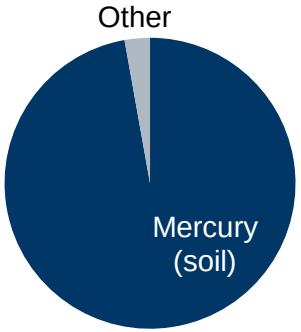
Human Toxicity Potential, Cancer

Contributing Toxins

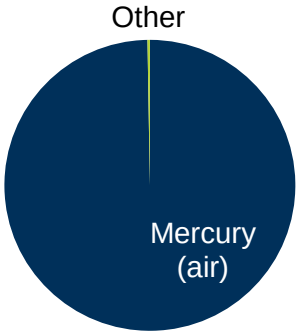
Landfill / Recycling



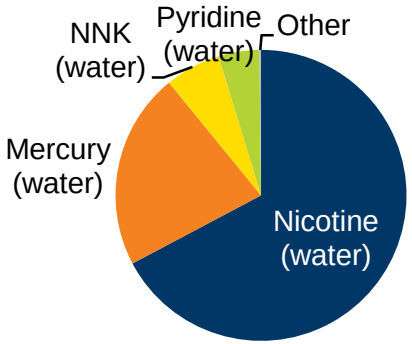
Unmanaged Waste (soil)



Incineration

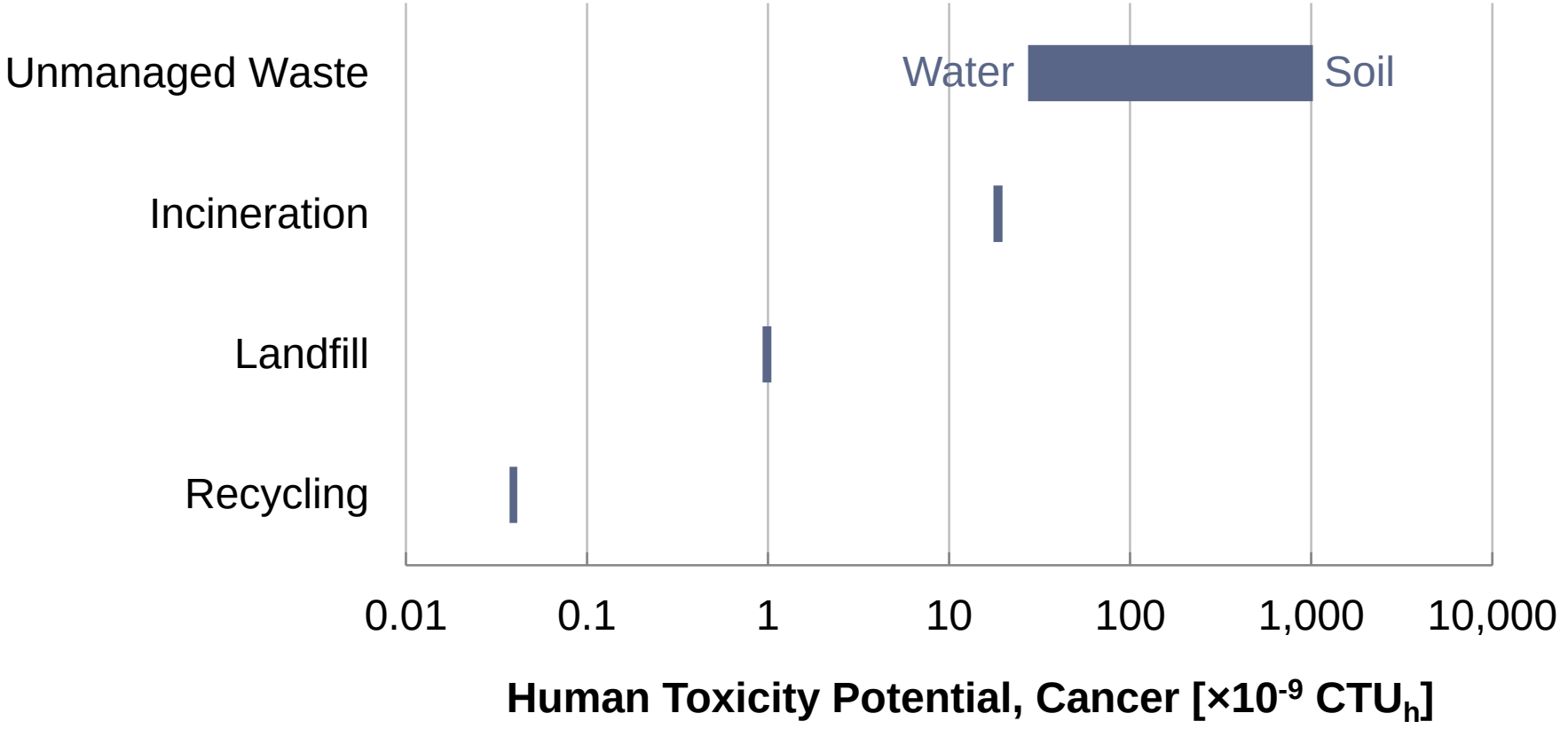


Unmanaged Waste (water)



Human Toxicity Potential, Cancer

Disposal System Impact Comparison



Summary

Interpretation and Conclusions

- LCA enables environmental performance assessment, including analysis of freshwater eco-toxicity and human toxicity aspects
- Cigarette waste drives toxicity impact—in particular nicotine and mercury content of tobacco and filter
- Unmanaged waste not associated with energy consumption; however…
  - Allows direct emission of toxins to environment
  - Leads to concerns such as aesthetics, marine debris, and fire hazards
- Managed waste reduces toxin emissions